ITA0329-MOBILE COMPUTING

ASSESSMENT TOOL – 2

Reg.No:192311411

Name: M.Nirosha

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Short Answers:

Q6. Channels per cell:

Given:

Spectrum = 42 MHz,

channel bandwidth = 300 kHz,

cluster size = 7

Total channels={42000}{300}=140

Channels per cell={140}{7} =20channels.

Q13. Received power in dBm:

Given:

Transmitted power = 8 W,

path loss = 33 dB

Pt(dBm)=10log10(8000) =39.03 dBm

Pr=39.03-33

6.03dBm.

Q8. Transmit power for doubling cell radius:

Given:

Radius doubles, inverse-square power law.

P R^2

Power factor=2^2=4

Therefore, transmit power must be increased 4 times.

Q14. Free-space path loss:

Given:

Frequency = 1900 MHz, distance = 2.5 km.

Formula:

FSPL=32.44+20log{10}(f)+20log{10}(d)

FSPL=32.44+20log{10}(1900)+20log{10}(2.5)

105.97 dB (approximately).

Q15. Total cluster call capacity:

Given:

6 cells, each supports 250 calls.

6*250=1500calls.

Q17. Total bandwidth consumed:

Given:

150 channels, each occupying 25 kHz.

150*25=3750kHz

3.75 MHz.

Q16. Maximum active calls:

Given:

4000 calls/second,

average call duration = 3

minutes = 180 seconds.

N=Call rate*Duration

N=4000*180

720000 active calls.

Q18. Total guard-band bandwidth:

Given:

30 adjacent frequency bands,

guard band = 40 kHz.

Assuming 29 gaps between 30 bands:

29*40=1160kHz=1.16MHz.

Q25. Total network user capacity:

Given:

1200 base stations, each supporting 600 users.

1200*600 =720000 users.

Q24. Maximum simultaneous users:

Given:

Bandwidth = 40 MHz,

bandwidth per user = 400 kHz.

{40000}{400} =100 users.

Q22. Number of FDMA channels:

Given:

Total spectrum = 6 MHz,

channel bandwidth = 30 kHz.

{6000}{30}=200channels.

Q23. Offered traffic in Erlangs:

Given:

1200 subscribers, 4 calls/hour,

average duration = 2.5 minutes.

A={1200*4*2.5}{60}

200 Erlangs.

Q21. Number of hexagonal cells:

Given:

Area = 400 km²,

radius = 400 m = 0.4 km

Using the specified approximation (A=R^2):

Acell=(0.4)^2=0.16 km^2

N={400}{0.16} ={2500 cells.

Q20. Total network user capacity:

Given:

8 regional centers,

40 base stations each,

250 users per station.

8*40*250 =80000 users.

Q19. Number of handoffs per hour:

Given:

Speed = 90 km/h,

cell radius = 1.5 km.

Cell diameter=2(1.5)=3 km

Handoffs={90}{3} =30 handoffs/hour.

Q1. Full-duplex voice channels:

Given:

Spectrum = 45 MHz,

simplex channel bandwidth = 25 kHz.

Full-duplex channels={45*1000}{2*25} =900 channels.

Q2. Channels allocated per cell:

Given:

Total channels = 560,

cluster size = 7.

Channels per cell={560}{7} =80 channels.

Q3. Total GSM voice channels:

Given:

100 carriers,

8 time slots per carrier.

100*8=800 voice channels.

Q5. Carrier-to-Interference Ratio (C/I):

Given:

Carrier = −60 dBm,

Interference = −92 dBm.

C/I=-60-(-92)

=32 dB.

Q4. Power control ratio in dB:

Given:

Base station power = 80 W,

handset power = 4 W.

10log{10}{80}{4} =10log{10}(20)

=13.01 dB.

Q7. Received power at twice the distance:

Given:

Initial power = 64 W,

path-loss exponent n=4.

P2={64}{2^4}={64}{16}

=4 W.

Q9. Total channels in a cluster:

Given:

Cluster size = 4,

channels per cell = 15.

4*15= 60 channels.

Q10. Doppler shift:

Given:

Velocity = 100 km/h,

frequency = 900 MHz.

v={100}{3.6}

=27.78 m/s.

Assuming maximum Doppler shift:

fd={vfc}{c} = {27.78*900*10^6}{3*10^8}

=83.33 Hz.

Q11. TDMA time-slot duration:

Given:

Frame duration = 5 ms,

number of slots = 8.

={5}{8}

=0.625 ms.